

Intracranial hypertension and herniation

CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPi < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations. ***Cushing's Triad is a LATE, pre-terminal sign of brainstem compression. Do not wait for its onset to initiate therapy.***

Syndrome	Anatomical Substrate	Clinical Exam & Diagnostic Trap
Uncal (Lateral)	Medial temporal lobe (uncus) pushed over tentorial edge	Ipsilateral sluggish/dilated pupil (CN III compressed), contralateral hemiparesis. <i>Trap: Kernohan's Notch</i> causes false-localizing ipsilateral hemiparesis.
Central (Axial)	Downward diencephalic and midbrain displacement	Progressive stupor, midpoint fixed pupils, decorticate to decerebrate posturing. <i>Trap: Symmetrical signs often confused with metabolic encephalopathy.</i>
Subfalcine (Cingulate)	Cingulate gyrus displaced under the falx cerebri	Often clinically silent, or presents with contralateral lower extremity weakness. <i>Trap: ACA compression</i> causes frontal/leg territory infarction.
Tonsillar (Downward)	Cerebellar tonsils forced through the foramen magnum	Cushing's triad, flaccid quadriplegia, respiratory arrest. <i>Trap: Cushing's triad is pre-terminal; do not wait for its appearance.</i>

MANAGEMENT (TIERED PATHWAY)

Tier 0: Fundamental Optimization (For all at-risk patients)

- Elevate HOB 30°; strict neutral midline neck alignment to preserve venous outflow.
- Euvolemia (isotonic saline; absolute ban on hypotonic $\$D_5W\$$), normothermia (treat fevers), normocapnia ($\$pCO_2\$$ 35–40 mmHg).

Tier 1: Initial Interventions

- Continuous CSF drainage via EVD (typically set at 10–15 cmH₂O).
- Analgesia/sedation (propofol/fentanyl) to prevent coughing, agitation, or ventilator dyssynchrony.

Tier 2: Hyperosmolar & Ventilatory Escalation

- **Mannitol 20%:** 1 g/kg IV bolus over 20–30 min. Must use in-line filter. **Hold if Serum Osmolarity > 320 mOsm/kg or Gap ≥ 20 mOsm/kg.**
- **Hypertonic Saline (HTS):** 3% (150–250 mL bolus) or 23.4% (30 mL rescue bolus; central line access only). **Hold if Serum Na > 155 mEq/L or Cl > 115 mEq/L.**
- **Controlled Hyperventilation:** Target $\$pCO_2\$$ 30–35 mmHg. Temporary bridge only; avoid prolonged use or $\$pCO_2 < 30\$$ (ischemia risk).
- **Steroids:** Contraindicated for cytotoxic edema in stroke.

Tier 3: Refractory & Salvage Measures

- High-dose barbiturate therapy (pentobarbital) titrated to burst suppression on EEG.
- **Decompressive Surgery:**
 - **Malignant MCA (DHC):** Age ≤ 60 years, GCS decline ≥ 1 / pupil changes, infarct $\geq 50\%$ MCA, within 48h (DECIMAL/DESTINY).
 - **Cerebellar Stroke:** Suboccipital craniectomy for brainstem compression, 4th ventricle effacement, or hydrocephalus.

TIER 0: BASE

hob 30°, midline, isotonic

TIER 1: INITIAL

evd drainage, sedation

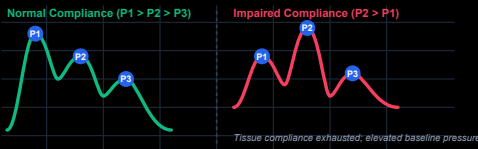
TIER 2: MEDICAL

osmotherapy, bridge vent

TIER 3: SALVAGE

barb coma, craniectomy

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
 - **Normal Compliance:** $P1 > P2 > P3$ (elastic brain tissue easily cushions pulsations).
 - **Impaired Compliance / High ICP:** $P2 > P1$ (brain tissue reserve exhausted; high risk of herniation).

CRITICAL WARNING: The Perfusion Trap ($\$CPP = MAP - ICP\$$)

In patients with mass effect, cerebral perfusion is highly pressure-dependent. Trainees must avoid the reflex to aggressively lower blood pressure. Lowering MAP will drop Cerebral Perfusion Pressure (CPP), causing immediate penumbral ischemia and expanding infarct volume. Maintain stroke blood pressure targets.